Multi String Search

AlgoExpert · Hard · Tries

The problem

- For each small string, does it occur in the big string?
- Trie of small strings, scanned from every start position of the big string.

Intuition

- Put the small strings in a trie, then walk it from each position of the big string.

Brute force vs optimal

- IndexOf per small string → $O(b \cdot \Sigma s)$
- Trie of smalls, walk from each index of big → $O(b \cdot \max S + \Sigma s)$
- Aho–Corasick → linear

Approach

- 1. Build a trie
- 2. Start a walk at each index of the big string
- 3. Mark words found
- 4. Return booleans

Worked example

```
this
yo
is
a
bigger
string
kappa
```

Build a trie of the small strings

Worked example

```
this is a big string

start at each index, walk the trie
```

Walk the trie from each start index

Worked example

small	found
this	✓
yo	×
is	✓
a	✓
bigger	×
string	✓
kappa	×

Results

Java solution

```java
public class MultiStringSearch {
    private static class Trie { Map<Character, Trie> next = new HashMap<>(); String word; }

    public static List<Boolean> multiStringSearch(String big, String[] smalls) {
        Trie root = new Trie();
        for (String s : smalls) {
            Trie t = root;
            for (char c : s.toCharArray()) t = t.next.computeIfAbsent(c, k -> new Trie());
            t.word = s;
        }
        Set<String> found = new HashSet<>();
        for (int i = 0; i < big.length(); i++) {
            Trie t = root;
            for (int j = i; j < big.length(); j++) {
                t = t.next.get(big.charAt(j));
                if (t == null) break;
                if (t.word != null) found.add(t.word);
            }
        }
        List<Boolean> out = new ArrayList<>();
        for (String s : smalls) out.add(found.contains(s));
        return out;
    }
```

Complexity

- Time: O(ns + bs) where b is the big length, s the longest small
- Space: O(ns)

Edge cases & pitfalls

- Overlapping matches
- Duplicate small strings
- Small longer than big

Interview tips & follow-ups

- Mention Aho–Corasick

Recap

- Put the small strings in a trie, then walk it from each position of the big string.
- Complexity: O(ns + bs) where b is the big length, s the longest small